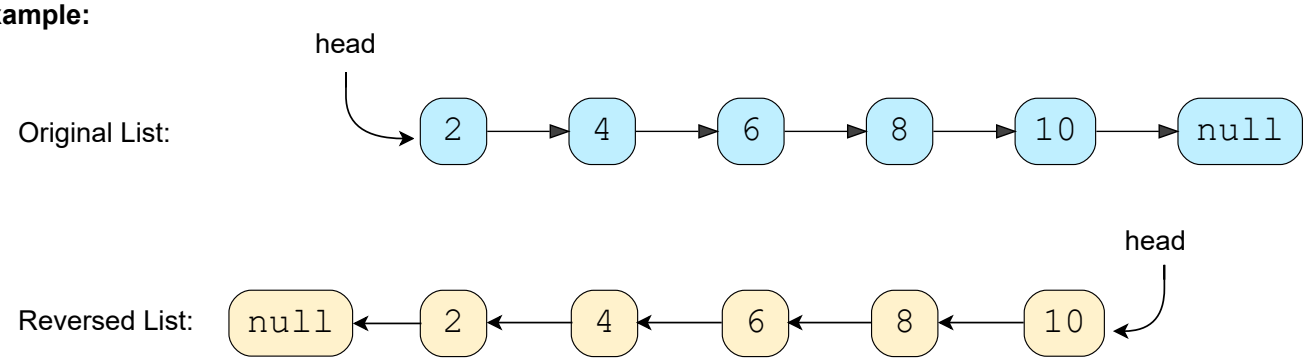
Solution: Reverse a LinkedList

Problem Statement

Given the head of a Singly LinkedList, reverse the LinkedList. Write a function to return the new head of the reversed LinkedList.

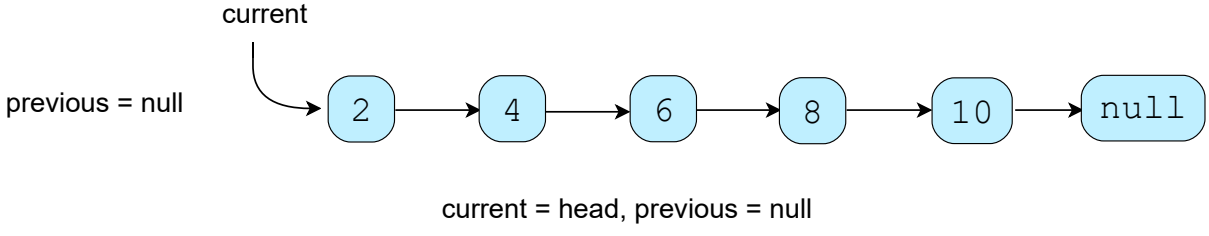
Example:

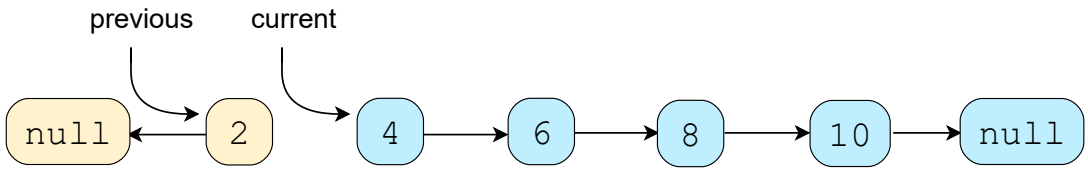


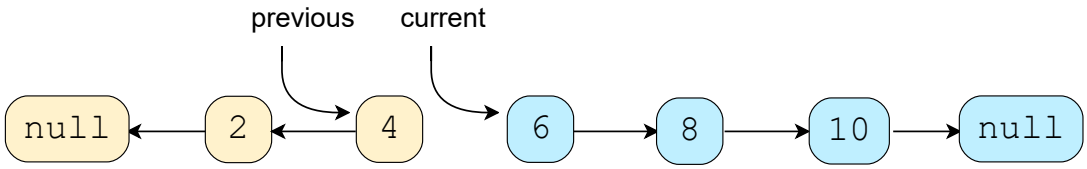
Solution

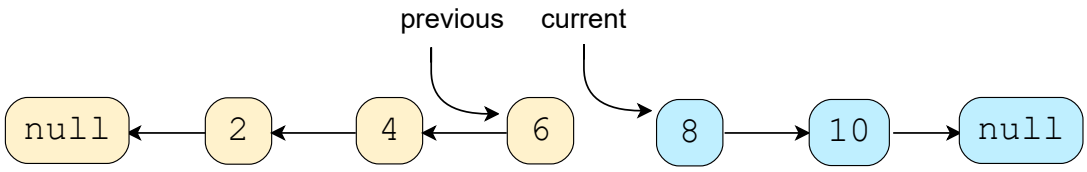
To reverse a LinkedList, we need to reverse one node at a time. We will start with a variable `current` which will initially point to the head of the LinkedList and a variable `previous` which will point to the previous node that we have processed; initially `previous` will point to null.

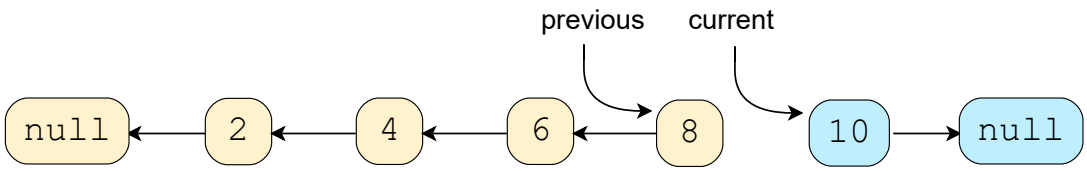
In a stepwise manner, we will reverse the `current` node by pointing it to the `previous` before moving on to the next node. Also, we will update the previous to always point to the `previous` node that we have processed. Here is the visual representation of our algorithm:

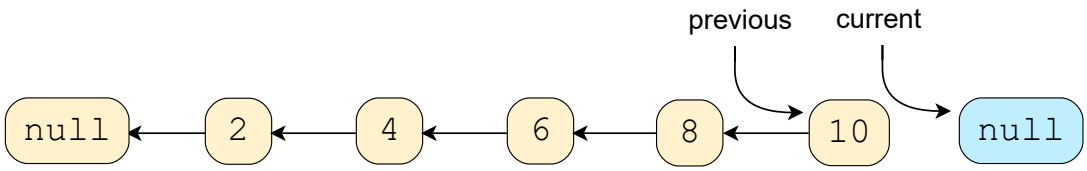


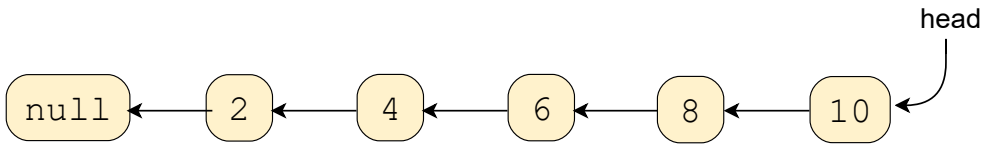

previous
current
null
2
4
6
8
10
null


previous
current
null
2
4
6
8
10
null


previous
current
null
2
4
6
8
10
null

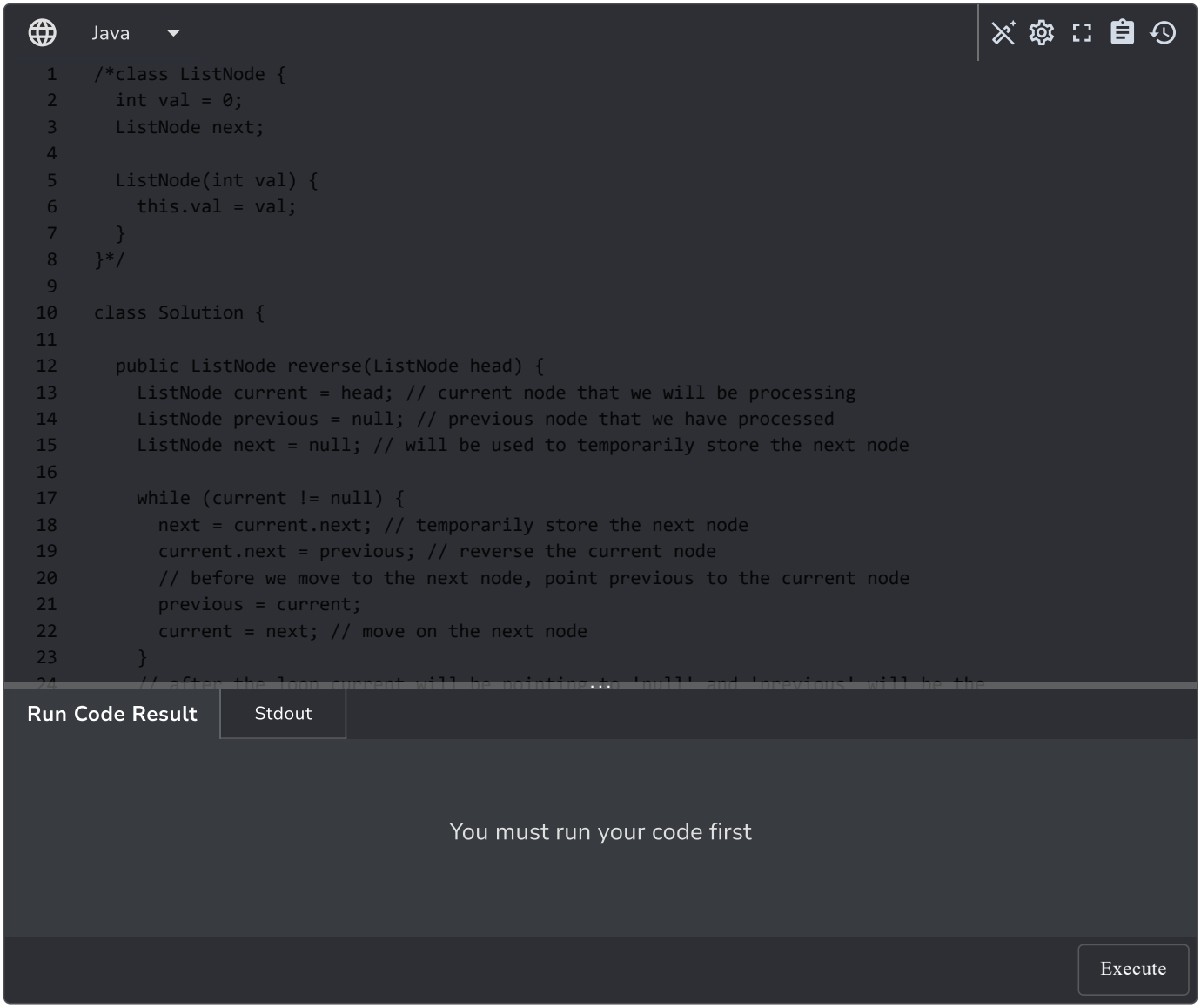

previous
current
null
2
4
6
8
10
null


previous
current
null
2
4
6
8
10
null


head
null
2
4
6
8
10

Code

Here is what our algorithm will look like:



Time Complexity

The time complexity of our algorithm will be O(N) where 'N' is the total number of nodes in the LinkedList.

Space Complexity

We only used constant space, therefore, the space complexity of our algorithm is O(1).

← Previous

Reverse A LinkedList (Easy)

Next →

Reverse A Sub-List (Medium)

Mark as Completed